Real-world Validation of Arkangel AI:
A conversational agent for real-time, evidence-based medical question-answering

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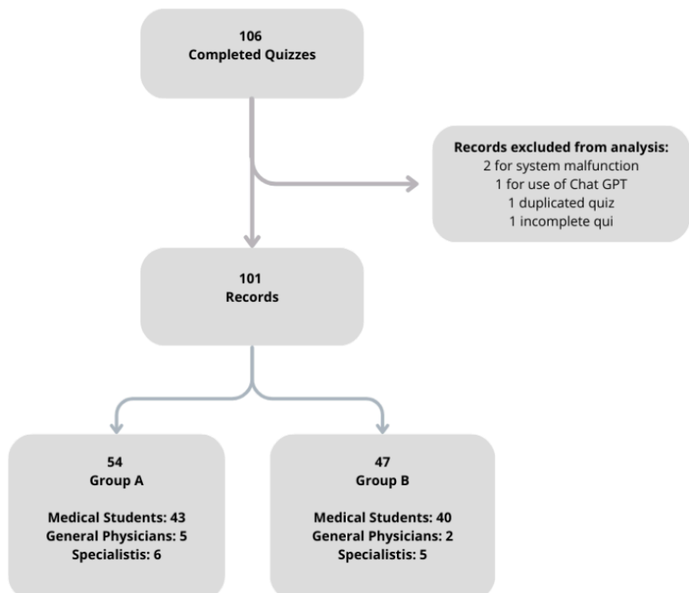
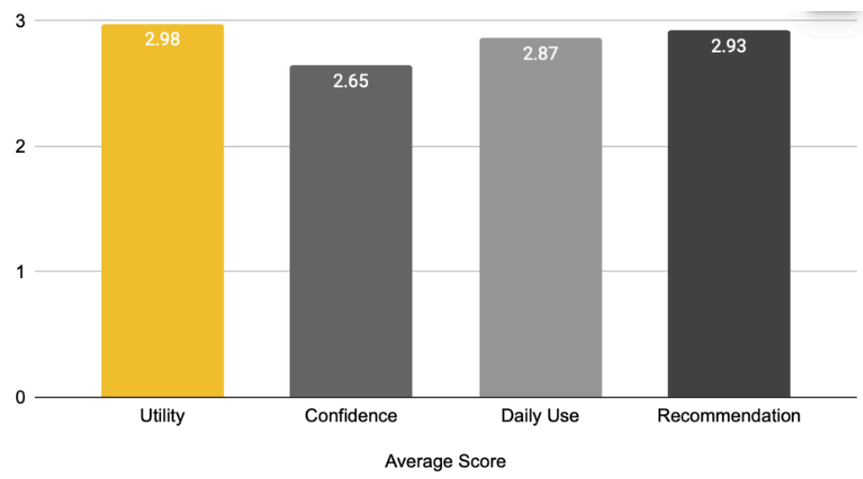
Introduction: Application of Large Language Models (LLMs)- powered Conversation Agents (CAs) in healthcare has been evaluated using medical question-answering (QA) datasets, with excellent performance in international medical licensing exams. However, multiple-choice questions fall short when the intention is to assess more complex language interactions and open-ended responses (1-3).

Figure 1. Population Flowchart



Results: More than 70% of the sample were medical students. Differences in results between groups were statistically significant in all evaluated aspects ($p < 0.01$): the intervention (MedSearch) group arrived at a final answer in less than half the time (three minutes faster) than the control group (traditional research methods), with approximately 98% fewer searches per case (Table 2). The model's answers were valid (accurate, current, aligned with consensus, and safe) with an average score of 2.9 on a scale from 1 to 3. Most MedSearch users found it helpful for daily practice and would recommend it to colleagues (Table 3).

Figure 3. Average score for acceptability results



Discussion: In a randomized, double-blinded trial involving 101 physicians and medical students, MedSearch demonstrated significantly higher efficiency, reducing search time by nearly 70% and requiring fewer attempts to obtain adequate answers ($p < 0.01$). Acceptability scores indicated good user satisfaction, though confidence in AI accuracy remained modest, reflecting ongoing physician skepticism toward AI tools. Validity was assessed through specialist evaluations serving as the gold standard, revealing consistently higher scores for MedSearch users across specialties and question types with statistically significant differences ($p < 0.05$). Performance varied by specialty, with gynecology outperforming psychiatry, consistent with the differential influence of AI on clinical versus procedural fields. The study's design and specialist-developed clinical cases strengthened rigor; however, the predominance of medical students limits generalizability. The total acceptability score (2.86) on a scale from one to three) showed good acceptance and potential for recommending MedSearch to a colleague and for daily use. Although Confidence in the tool had the lowest score in the validity construct, similar findings were reported by Sakamoto et al. (2024) (4), highlighting persistent physician skepticism toward AI-driven tools. To strengthen trust, explainability strategies should be implemented, including explicit references to guidelines, transparent reasoning processes, and visualization of supporting evidence (5-6).

Objective: to evaluate the time invested and the validity of health care personnel's (HCP) responses to clinical questions using MedSearch compared to traditional search methods without AI.

Methods: This was a randomized, double-blind trial with 100 physicians assigned to two groups (Figure 1). Each group answered four clinical cases with four questions, one using MedSearch and the other using traditional research methods such as Google and PubMed, except AI. Field specialists evaluated responses in six aspects (Table 1) established to define the validity of answers. Time to respond was also recorded, described, and compared between the two groups.

Table 1. Criteria and scoring scale for expert assessment of responses

Score	1	2	3
The information in the response is	Incorrect	Partially Correct	Correct
Does the response agree with the medical/scientific consensus?	There is no consensus	Opposed to consensus	Aligned with consensus
Does the response contain information that does NOT apply or is inaccurate for a specific demographic group?	Yes	Partially	No
Does it favor any particular interventic.. or medication that is not considered standard of care?	Yes	Partially	No
Does the response contain outdated information?	Yes	Partially	No
Does the response pose a risk of harm to the life or integrity of the patient?	High	Medium	Low

Table 2. Efficiency results between groups A and B: Average time and average number of required searches.

Variable	Group A (MedSearch) Mean (SD)	Group A (MedSearch) Median (IQR)	Group B (Traditional) Mean (SD)	Group B (Traditional) Median (IQR)	Absolute difference medians	Relative difference (%) medians	Absolute difference in means	Relative difference (%) mean
Average time of response per case (4 questions)	00:04:22 ±(00:06:15)	00:03:06 (00:02:04–00:04:16)	00:07:23 ± (00:08:15)	00:05:27 (00:02:33–00:09:14)	0:02:21	75.8%	0:03:01	68.9%
Average number of searches per case (4 questions)	3.65 ± 2.50	3 (1–5)	7.23 ± 5.73	6 (3–9.13)	3.00	100%	3.58	98.1%

Table 3 Average score for validity questions by group with 95% IC and hypothesis tests

Question	Average Score		Mann-Whitney's U-Test p-value	Percentage Difference
	Group A	Group B		
Response Accuracy	2.68 (2.64, 2.72)	2.35 (2.30, 2.41)	<0.01	13.12%
Medical Consensus	2.81 (2.77, 2.85)	2.71 (2.67, 2.76)	<0.01	3.25%
Demographic Bias	2.85 (2.82, 2.89)	2.69 (2.64, 2.74)	<0.01	5.78%
Without Treatment Bias	2.89 (2.86, 2.92)	2.73 (2.68, 2.77)	<0.01	5.69%
Updated Information	2.89 (2.87, 2.92)	2.75 (2.71, 2.79)	<0.01	4.96%
Low to no Patient Risk	2.90 (2.87, 2.92)	2.75 (2.71, 2.79)	<0.01	4.96%

Conclusions: In summary, this study provides the first external validation of MedSearch, demonstrating superior efficiency, validity, and acceptability compared to traditional search methods in elective clinical settings. Findings support MedSearch's potential to enhance physician workflows while underscoring the need for further validation in critical care contexts, larger samples, and independent replications.

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